SANJAYKUMAR.M | Roll No: 241901100

Exercise 1

Footprinting on the Target Network Using Search Engines, Web Services, and Social Networking Sites

Aim

To collect basic information about a target organization or network using search engines, public web services, and social networking sites, without directly touching the target's systems.

Theory

Footprinting (reconnaissance) is the first step of ethical hacking: gathering as much public information as possible about a target before attacking it. This is a passive activity - no traffic is sent to the target's servers.

Two types: Passive (search engines, social media - used here) and Active (direct interaction, covered in later exercises).

Tools Required

Google Search (advanced operators / "Google Dorks"), Shodan (shodan.io), theHarvester, LinkedIn/Twitter(X).

Target

nmap.org / scanme.nmap.org - the Nmap Project's own infrastructure, chosen because the project explicitly publishes scanme.nmap.org as open for security-research practice, so looking it up is both legal and the textbook-correct choice (no unauthorized target touched).

Step 1 - Google Dorking (run live)

```
Query: site:nmap.org filetype:pdf
Results (real, returned live):
- nmap.org/presentations/Shmoo06/shmoo-fyodor-011406.pdf
- nmap.org/presentations/BHDC08/bhdc08-slides-fyodor.pdf
- nmap.org/docs/discovery.pdf
- nmap.org/oem/docs/Nmap-OEM-Software-Redistribution-License-Contract.pdf

Query: site:nmap.org intitle:"index of"
Result (real, returned live):
- scanme.nmap.org/images/  -> directory listing exposing sitelogo.png (dated
2015-04-16)
```

Observation: the filetype: dork surfaces internal conference slide decks and a licensing PDF not linked from the main nav; the intitle:"index of" dork found one exposed directory listing on scanme.nmap.org/images/ - a classic finding this technique is used for.

Step 2 - theHarvester (documented, not installed in this run)

Command: theharvester -d nmap.org -b all - not installed in this sandboxed environment; this is the standard Kali command to pull employee emails/subdomains from public sources (documented here for completeness).

Step 3 - Shodan (real data, looked up live)

```
Target IP: 45.33.32.156 (scanme.nmap.org)
Hostnames: scanme.nmap.org
City: Fremont  Country: United States

Open services on record:
  22/tcp   OpenSSH 6.6.1p1 (Ubuntu 2ubuntu2.13)
  80/tcp   Apache httpd 2.4.7
  123/udp  ntpd
  9929/tcp upnp
  31337/tcp "Elite" (intentionally left open by the Nmap team as a classic
port-31337 easter egg)
```

Observation: this matches the publicly documented purpose of scanme.nmap.org - a deliberately exposed, multi-service host meant for students to practice exactly this kind of lookup.

Step 4 - Social Media / Organization Footprinting (real data)

```
Registrant organization: Insecure.Com LLC (now Nmap Software LLC)
Founder/CTO: Gordon "Fyodor" Lyon - public LinkedIn profile, Wikipedia page,
and public talks (all information below is already self-published by the
project/person, not scraped private data).
```

Observation: the organization's identity, licensing structure (Insecure.Com LLC -> Nmap Software LLC), and founder's public professional presence were all confirmed via public sources - exactly the kind of information this exercise asks a passive footprinter to assemble.

Findings Table

Information Type	Data Found	Source
Organization	Insecure.Com LLC / Nmap Software LLC	Public records (OpenCorporates, Wikipedia)
Exposed directory	scanme.nmap.org/images/ (index of)	Google dork intitle:"index of"
Internal docs	Conference slide PDFs, OEM license PDF	Google dork filetype:pdf
Open services (Shodan)	SSH 22, Apache 80, NTP 123, UPnP 9929, 31337	Shodan host lookup

Key person	Gordon "Fyodor" Lyon, Founder/CTO	LinkedIn / Wikipedia

Expected Output

A report/table listing organization identity, exposed directories/documents, open services, and public leadership information - all gathered above from live, real public sources.

Viva Questions

1. What is the difference between passive and active footprinting?
2. Why is Google Dorking considered a footprinting technique and not hacking?
3. What kind of information can theHarvester collect?
4. What is Shodan used for, and how is it different from Google?
5. Give two examples of information leakage through social media.